

AURA – Universal Assistant for Response and Assistance for the Elderly

Artificial Intelligence of Things

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Abstract—This paper presents the design and implementation of AURA, an embedded Artificial Intelligence of Things (AIoT) system designed to assist elderly people living independently. The proposed solution integrates sensors, machine learning models, and embedded hardware to monitor user activity and provide assistance through simple voice interaction.

The system is based on two lightweight TinyML models developed using Edge Impulse and deployed on an ESP32-S3 microcontroller. The first model implements keyword spotting (KWS) to detect a predefined wake word ("AURA") through audio signals captured by a microphone. The second model performs activity recognition using visual input to identify movement patterns and potential abnormal situations.

Both models are designed to run directly on the embedded device using Edge AI techniques, reducing latency, improving privacy, and enabling real-time response without relying entirely on cloud processing. Experimental validation demonstrates the feasibility of integrating voice detection and activity recognition in a low-power AIoT architecture aimed at improving safety and assistance for elderly users.

Index Terms—Artificial Intelligence of Things (AIoT), Edge AI, Keyword Spotting, Activity Recognition, Embedded Systems, Edge Impulse, Elderly Assistance

I. INTRODUCTION

Population aging has become one of the most relevant social challenges of the coming decades [1]. More and more elderly people live alone or in conditions of partial dependency, which increases the need for solutions that simultaneously contribute to their safety at home, assistance in daily activities, and support in situations of isolation [1]. In many cases, the absence of monitoring mechanisms can generate significant risks, especially in the face of events such as falls, prolonged periods of inactivity, or domestic emergencies that require a rapid response.

At the same time, technology is increasingly becoming part of the everyday environment. However, a large proportion of elderly people face what is known as the digital divide, that is, difficulties interacting with devices and technological platforms designed without considering their cognitive, physical, or usability needs [2]. Traditional interfaces are often complex or not very intuitive, limiting user autonomy and delaying the

request for help in critical situations. This issue highlights the need for technological systems that operate in a more natural way, with minimal user interaction and autonomous operation capabilities.

In this context, Artificial Intelligence of Things (AIoT) systems represent a promising alternative to address these challenges. The integration of sensors, embedded devices, and machine learning models makes it possible to perceive the environment, analyze behavior patterns, and generate real-time responses. In particular, the use of edge computing (Edge AI) enables artificial intelligence models to run directly on low-power devices, reducing latency, improving data privacy, and allowing operation even in scenarios with limited connectivity [4].

Based on this approach, this work presents AURA (Universal Assistant for Response and Assistance for the Elderly), an embedded AIoT system designed to support the well-being and safety of elderly people in domestic environments. The proposal integrates sensors for environmental perception, machine learning models trained on the Edge Impulse platform, and feedback mechanisms through actuators and IoT services. The models are deployed on an ESP32-S3 microcontroller and follow an edge-first strategy, where processing is carried out locally to guarantee low latency and greater data privacy. In this way, the system demonstrates how the combination of AIoT, TinyML, and embedded systems can contribute to the development of accessible intelligent assistants aimed at improving the quality of life of elderly people.

II. PROPOSED SOLUTION

A. General System Architecture

The AURA system is based on a distributed AIoT architecture that allows environmental information to be captured, processed locally, and transformed into real-time responses or notifications. As shown in Figure ??, the system operation is organized into three main layers: perception, edge processing, and actuation.

The perception layer is responsible for capturing signals from the environment and the user through sensors integrated

into the device. In the developed system, a microphone is mainly used to register voice commands or activation words, and a camera captures visual information related to the user's activity or posture.

These signals are then sent to the edge processing layer, which runs directly on the ESP32-S3 microcontroller. At this stage, the initial signal processing is carried out, as well as the execution of the machine learning models developed on the Edge Impulse platform. Local processing allows real-time inference without depending on cloud services, reducing latency and communication resource consumption.

Finally, the actuation layer is responsible for generating the system's response once the artificial intelligence models produce a prediction. Depending on the detected event, the system can emit sound alerts, generate voice responses, or register events within the IoT platform for later visualization by family members or caregivers. This integration allows the system not only to interact with the user, but also to facilitate remote monitoring in possible risk situations.

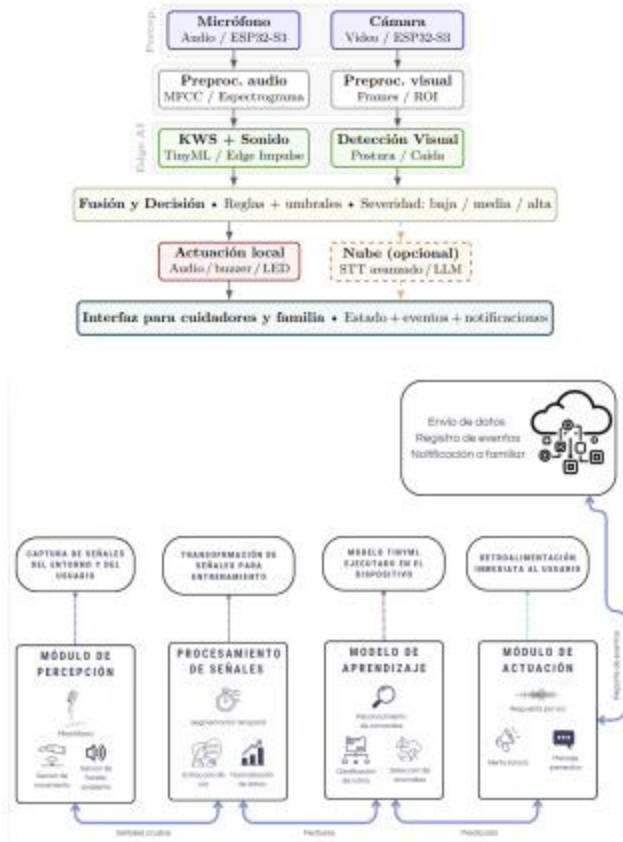


Fig. 1. General architecture of the AURA system and processing flow from signal capture to the generation of responses and notifications.

B. Operation Flow

The general operation of the system follows a processing sequence composed of several stages. First, the sensors integrated into the ESP32-S3 Sense device capture information from the user's environment. In this case, the microphone and the camera are mainly used, which make it possible to obtain audio and video signals representing what is happening around the elderly person. These signals are sent to the system to be analyzed.

Subsequently, the processed data are analyzed by two machine learning models that fulfill different functions within the system. The first corresponds to the keyword recognition model, whose objective is to detect when the user pronounces the assistant's activation word, in this case the word "AURA". When the system identifies this word, the assistant is automatically activated and begins recording a short audio fragment corresponding to the user's request, thus allowing the action or help being requested to be interpreted.

The second model corresponds to movement recognition, which uses the information captured by the camera to analyze the user's posture or activity. This model was trained to recognize mainly two states: *standing* (when the person is standing up) and *lying* (when the person is lying down). The identification of these states allows the system to detect potentially dangerous situations, such as a possible fall or accident, especially if the system identifies that the person remains on the floor for a prolonged period of time.

Finally, once the models make their predictions, the system applies a series of decision rules that determine the action to be executed. Depending on the obtained result, the system can generate alerts, activate sound signals, or register the event within the IoT platform. This information can later be viewed by a family member or caregiver through the system's web interface, allowing monitoring of the user's condition and enabling a rapid response to a possible emergency.



Fig. 2. Interaction with the web page by the user or the person responsible for the elderly person

III. IMPLEMENTATION DETAILS

A. Dataset Construction

The development of the system required the construction of two independent datasets corresponding to the two machine learning models implemented.

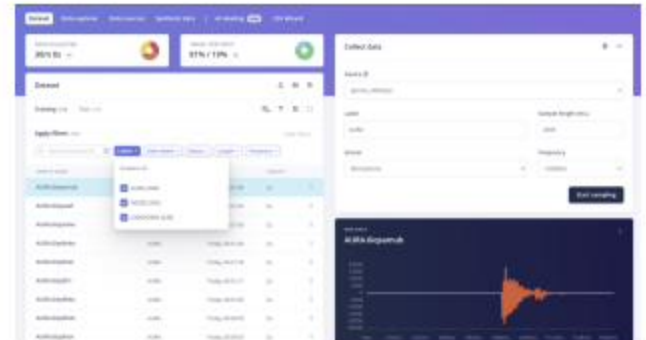


Fig. 3. Data collected using Edge Impulse for the voice recognition model

For the keyword recognition model (KWS), audio samples were collected that include the target word "AURA"

pronounced by the user under different environmental noise conditions, as well as background noise samples and unrelated sounds that improve the model's discrimination capability, labeled as AURA, NOISE, and UNKNOWN respectively, taking around 300 samples for each one.

For the movement recognition model, a public-domain dataset called "Le2i Fall Dataset" [4] was used. Additionally, video sequences were captured representing different activities such as standing or lying on the floor. Around 1,300 samples were collected. These samples were labeled according to the type of observed movement with the aim of training a model capable of recognizing basic behavior patterns.



Fig. 4. Data collected for the movement recognition model in Edge Impulse and a sample of the data

B. Data Preprocessing

Data preprocessing was performed independently for each of the machine learning models implemented in the system: the keyword recognition model (Keyword Spotting) and the vision-based movement recognition model.

For the keyword recognition model, audio signals captured through a microphone were used. The signal was processed as time-series data with a sampling frequency of 16 kHz, which makes it possible to adequately capture the relevant characteristics of the human voice. During the segmentation process, 1000 ms time windows with a 500 ms stride between consecutive windows were used. This overlap makes it possible to generate multiple samples from the same signal and improves the model's ability to detect the keyword at different positions within the audio. Additionally, zero padding was applied to maintain a consistent size for the input windows. Subsequently, the signals were transformed using the Mel Frequency Energy (MFE) method, which converts the acoustic signal into a spectral representation that facilitates learning the characteristic patterns of the keyword.

In the case of the movement recognition model, preprocessing was carried out on visual data obtained through the device camera. The images were resized to a resolution of 96x96 pixels in order to reduce computational load and allow the model to run on the embedded device. To preserve the original image proportion, a resizing method based on *fit shortest axis* was used. This procedure makes it possible to maintain the relevant spatial characteristics of the scene while generating a consistent input for the machine learning model. The processed images were later used to train an object detection model capable of classifying two main user states: standing position (*standing*) and lying position (*lying*).

The preprocessing applied to both models makes it possible to transform the data captured by the sensors into structured representations suitable for the training and inference of artificial intelligence models on embedded devices.

C. Feature Extraction

During this stage, relevant features were extracted from the processed data. In the voice recognition model, spectral representations of the audio signal were used, such as spectrograms or cepstral coefficients (MFCC), which make it possible to capture acoustic patterns characteristic of the keyword.

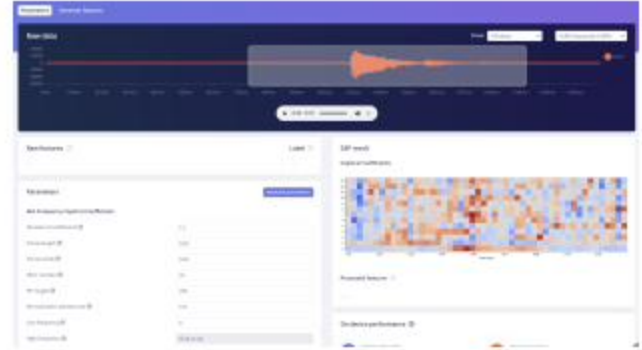


Fig. 5. Feature extraction section of the voice recognition model

In the movement recognition model, visual features derived from the images captured by the camera were used, allowing spatial patterns associated with user movement within the environment to be identified.

D. AI Model Architecture

The artificial intelligence models used in the AURA system were developed using the Edge Impulse platform, which makes it possible to design and train optimized models for embedded devices through TinyML techniques.

The first model corresponds to keyword recognition (*Keyword Spotting*), which analyzes audio signals captured by the microphone with the objective of detecting the activation word "AURA". When this word is identified, the system activates the assistant and allows interaction with the user to begin. This model makes it possible to distinguish between the keyword, environmental sounds, and other audio signals.

The second model corresponds to movement recognition, which uses images captured by the camera to analyze the user's posture. This model was trained to identify mainly two states: *standing*, when the person is standing, and *lying*, when the person is lying on the floor. This information allows the system to detect potentially dangerous situations, such as a possible fall.

Both models were optimized to run directly on the ESP32-S3 microcontroller, making it possible to perform real-time inference on the device. This allows the system to respond quickly to detected events and reduces dependency on external processing.

E. Training and Validation

The models were trained using the Edge Impulse platform, which makes it possible to develop and optimize machine learning models oriented toward embedded systems.

In the case of the keyword recognition model (Keyword Spotting), a neural network trained from acoustic features obtained through Mel-frequency cepstral coefficients (MFCC) was used. The model was trained with audio samples corresponding to three classes: *AURA*, *NOISE*, and *UNKNOWN*. During the training process, the model parameters were adjusted with the objective of improving its ability to distinguish the keyword from other environmental sounds.

For movement recognition, the FOMO (Faster Objects, More Objects) algorithm was used, based on MobileNetV2, which is designed for object detection tasks on devices with limited resources. This model makes it possible to analyze images captured by the camera and identify different user states. In this case, the system was trained to recognize two main classes: *lying* and *standing*, in order to detect possible relevant situations related to user activity.

Once training was completed, the models were evaluated using validation sets that made it possible to measure their performance through metrics such as precision, recall, and F1-score. The obtained results show that both models achieve good classification capability. Finally, the models were optimized and exported for direct execution on the embedded device, making it possible to perform real-time inference within the AURA system.

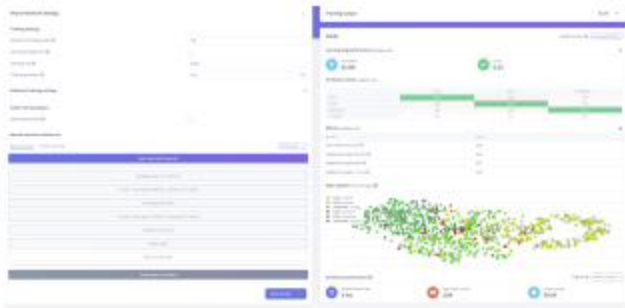


Fig. 6. Training data and performance of the AURA word recognition model and other sounds

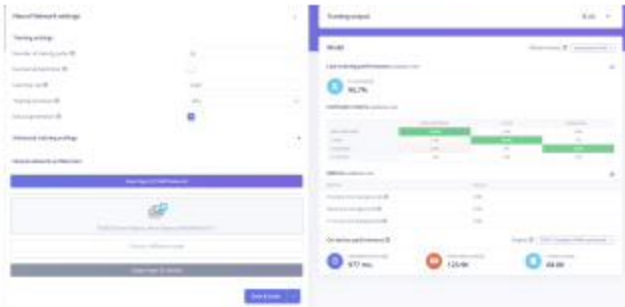


Fig. 7. Training data of the movement recognition model

F. Model Testing

Once the training process was completed, an evaluation stage was carried out using the *Model Testing* tool provided by the Edge Impulse platform. This functionality makes it possible to analyze the behavior of the model using test data that were not used during training, in order to measure its generalization capability and performance in new scenarios.

For the keyword recognition model (KWS), labeled audio samples within the test dataset were used. Each sample was processed by the model and compared with its expected label. The obtained results show an overall accuracy of 87.37%, which indicates that the model is capable of correctly identifying the activation word “AURA” in most cases. Additionally, metrics such as *precision*, *recall*, and *F1-score*, as well as the confusion matrix, were analyzed, which made it possible to observe how predictions are distributed among the *AURA*, *NOISE*, and *UNKNOWN* classes. These results show that the model adequately differentiates between the keyword and environmental sounds.

On the other hand, for the movement recognition model, a similar evaluation process was carried out using test data corresponding to different user postures. In this case, the model was trained to identify the two states: *standing* and *lying*. The results obtained during the *Model Testing* stage show an overall accuracy of 91.25%, with *precision* and *recall* values close to 0.95. This indicates that the model is capable of correctly recognizing the user’s postures in most of the evaluated situations.

Overall, the results obtained during the testing phase show that both models present adequate performance for their implementation in the AURA system. These evaluations validate that the models can be used on the embedded device to perform real-time inference, contributing to user monitoring and the detection of possible risk situations.

G. System Deployment

Once the models were trained, they were exported from the Edge Impulse platform and deployed on the embedded device through the Arduino library. To improve execution efficiency, the EON compiler and the quantized version of the model (*int8*) were used, which makes it possible to reduce memory usage and optimize performance on resource-constrained devices.

Because the AURA system integrates two machine learning models, one for keyword recognition and the other for movement recognition, their simultaneous execution on the ESP32 microcontroller represented a challenge in terms of memory consumption and processing. To achieve the execution of both models within the device, it was necessary to use a specific module generated by Edge Impulse that refers to the combination allowing multiple impulses to be integrated within the same project, optimizing the use of system resources and allowing both models to run on the same device; more specifically, this is called a multiple impulses combination [5].

The performance metrics obtained during the deployment process show that both models can run directly on the selected

microcontroller, maintaining adequate inference times and memory consumption compatible with the device capabilities. In this way, the system is capable of performing real-time inference without depending on external processing, which is fundamental for applications based on AIoT and Edge AI.

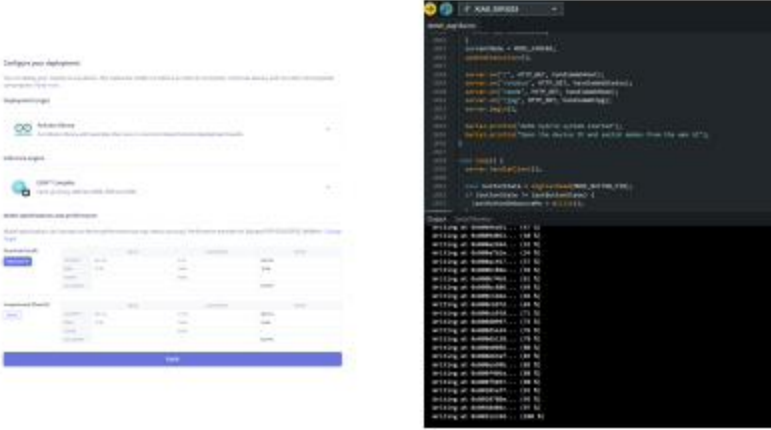


Fig. 8. System deployment in Edge Impulse and on the ESP32-S3

IV. RESULTS AND ANALYSIS

Once the trained models were integrated into the embedded system, functional tests were carried out with the objective of evaluating the behavior of the complete AURA system in a real environment. The experimental results demonstrate that both models present adequate performance for their implementation on low-power embedded devices. In particular, the keyword recognition model (Keyword Spotting) achieved an approximate accuracy of 87.37%, while the posture recognition model obtained an accuracy close to 91.25% during the testing stage. These results show that it is possible to run lightweight machine learning models directly on microcontrollers such as the ESP32-S3 while maintaining real-time inference capabilities.

As part of the IoT implementation of the system, a web interface was developed that allows real-time visualization of the system state and the events detected by the device. Through this page, indicators such as the user's general status, fall detection, current movement, and recent activity are presented. When the system detects a potentially dangerous situation, such as a possible fall or an anomaly in the user's activity, the interface generates an alert that can be viewed by a family member or caregiver of the elderly person. This allows a responsible person to take timely action in the event of a possible emergency, thus facilitating remote user monitoring.

Additionally, the keyword recognition module was implemented to activate the assistant through voice interaction. During the tests, the system was able to correctly identify the defined keyword to activate the assistant. Once the word is detected, the system automatically starts recording an audio fragment for a few seconds, which corresponds to the user's command or request. This audio can later be processed to interpret the user's intention and generate an appropriate assistant response.

The tests performed show that both modules work in an integrated way within the embedded system, making it possible to combine physical event detection through vision with



Fig. 9. Web page developed for interaction with the person responsible for the elderly person



Fig. 10. Demonstration of the recognition model results through serial terminal and web page

voice interaction. This integration demonstrates the feasibility of using TinyML and Edge AI techniques on low-power devices to develop intelligent assistants aimed at monitoring and supporting elderly people in domestic environments.

V. LIMITATIONS

Despite the obtained results, the system presents some limitations that must be considered. First, the keyword recognition model was trained mainly with the voice of a single group member, which may affect its generalization capability when it is used by people with different vocal characteristics.

Likewise, the quality of the sensors used in the prototype, particularly the camera and microphone integrated into the embedded system, may influence the accuracy of activity and voice recognition since they do not have very good range. Factors such as environmental noise, lighting variations, or changes in the user's position may impact model performance.

Another relevant limitation is related to the hardware restrictions of the microcontroller used. The memory and processing capacity of the ESP32 requires the use of optimized and reduced-size models, which limits the complexity of the algorithms that can be executed simultaneously on the device.

Finally, the system currently recognizes a limited set of user activities, mainly related to basic body posture (for example, standing or lying down). This restricts the system's ability to interpret a greater variety of behaviors or situations within the domestic environment.

VI. CONCLUSIONS

The development of AURA demonstrated that it is possible to implement artificial intelligence models in embedded systems using TinyML techniques, achieving direct inference execution on a microcontroller with limited resources. This highlights the potential of AIoT solutions for real-time intelligent monitoring applications.

Likewise, the project confirms that intelligent assistants can adapt to the specific needs of users, particularly in contexts of support for elderly people. The integration of sensors, machine learning models, and IoT platforms makes it possible to generate systems capable of detecting relevant events and communicating this information to family members or caregivers in a timely manner.

During the development, it was also identified that the quality and diversity of the dataset used for training directly influence the performance of recognition models, especially in tasks such as keyword detection. This highlights the importance of having representative data to improve the robustness and generalization of the system.

Finally, the integration of sensors and artificial intelligence models within an embedded system demonstrates that it is possible to develop technological solutions with social impact, aimed at improving the monitoring and safety of people within the home.

VII. FUTURE WORK

As future work, it is proposed to expand the dataset used for training the voice recognition model by incorporating samples from multiple users in order to improve the robustness and generalization capability of the system.

Likewise, it is proposed to integrate additional sensors or wearable devices that allow more complete information to be obtained about the user's condition, such as accelerometers, biometric sensors, or health monitoring devices.

Another line of improvement consists of optimizing and expanding the machine learning models used, with the objective of recognizing a greater variety of activities and improving the system's accuracy under different environmental conditions.

Finally, the integration of the system with smart home devices within a broader IoT ecosystem is considered, which would make it possible to automate responses to detected events and improve the user's assistance capability. In addition, future implementations could evaluate the system in real environments with multiple users to validate its performance in everyday usage scenarios.

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